



NEW DISEASE REPORT

First report of maize streak virus affecting maize in the Philippines

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corn virus diseases

Corn (maize; *Zea mays*) is the second most economically important crop in the Philippines. White corn is the main staple of some Filipinos, while yellow corn is used as livestock feed. Corn is also used as a raw material for industrial and value-added products. The major corn-growing areas in the country are Region I (Ilocos Region) and Region II (Cagayan Valley) in Luzon Island and Region X (Northern Mindanao), Bangsamoro Autonomous Region, and Region XII (Soccsksargen) in Mindanao (Philippine Statistics Authority, 2023).

Maize streak, caused by maize streak virus (MSV), a member of the species *Mastrevirus storeyi* (genus *Mastrevirus*, family *Geminiviridae*), is one of the serious constraints affecting corn production in many African and some Asian countries (Centre for Agriculture and Bioscience International, 2021). It has a monopartite single-stranded DNA genome of about 2,700 nucleotides encapsidated in icosahedral geminate particles (Fiallo-Olivé et al., 2021). The leafhopper, *Cicadulina mbila*, and other *Cicadulina* spp. transmit the virus persistently.

Corn with typical symptoms of maize streak disease (Fig. 1) was observed in four municipalities in Bukidnon (Region X) and three municipalities in South Cotabato (Region XII) during field surveillance in June and August 2023. Leaf samples from 28 diseased plants were collected across the seven municipalities and tested for the presence of MSV using PCR with degenerate primers (Tembo et al., 2020). MSV was detected in all seven municipalities and in 21 of the samples. Control samples from three healthy corn plants were negative for MSV. Amplicons (c. 1300 bp) of one isolate from Bukidnon (Manolo For-



FIGURE 1 Maize streak virus disease symptoms: a) round scattered spots, which become longer streaks and eventually coalesce and appear more profuse at the base of the initially infected leaf; and b) narrow, broken, or elongated, pale yellow streaks along the veins on dark-green foliage develop on the leaves that emerge later.

tich) and one from South Cotabato (General Santos City) were Sanger sequenced in both directions, and the sequences were assembled using Geneious Prime® Version 2023.2.1. When the sequences were aligned and compared using BLASTn, the isolates had >99% nucleotide identity. BLASTn analysis also showed that the isolates had highest identity (98.68%) with the MSV-A strains from Zimbabwe (AJ012634.1) and South Africa (Y00514.1). The close relationship between the Philippine isolates and the strains from Zimbabwe and South Africa was

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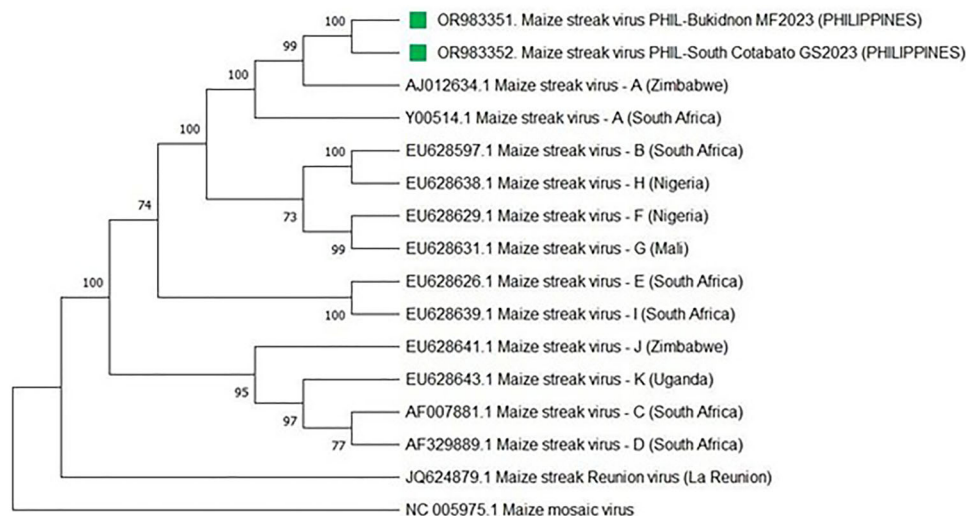


FIGURE 2 Phylogenetic tree based on partial nucleotide sequences including the replication-associated and partial movement genes, and long intergenic region of maize streak virus (MSV) isolates from the Philippines (OR983351: MSV PHIL-Bukidnon MF2023; and OR983352: MSV PHIL-South Cotabato GS2023) with reference sequences from GeneBank (AJ012634.1: MSV-A (Zimbabwe); Y00514.1: MSV-A (South Africa); EU628597.1: MSV-B (South Africa); AF007881.1: MSV-C (South Africa); AF329889.1: MSV-D (South Africa); EU628626.1: MSV-E (South Africa); EU628629.1: MSV-F (Nigeria); EU628631.1: MSV-G (Mali); EU628638.1: MSV-H (Nigeria); EU628639.1: MSV-I (South Africa); EU628641.1: MSV-J (Zimbabwe); EU628643.1: MSV-K (Uganda); and JQ624879.1: maize streak Reunion virus (La Reunion)). Sequences were aligned using ClustalW and the tree was constructed using the neighbour-joining method with 1000 bootstrap replicates. Maize mosaic virus (NC00575.1) was used as an outgroup.

supported by phylogenetic analysis performed using MEGA v. 11.0.10 (Fig. 2). Thus, the Bukidnon and South Cotabato isolates were identified as MSV, and the isolates were named PHIL-Bukidnon MF2023 (GenBank Accession No. OR983351.1) and PHIL-South Cotabato GS2023 (OR983352.1), respectively.

This report presents the first detection of MSV affecting corn in the Philippines. The virus isolates will be further characterised by whole genome sequencing. Field surveillance is being conducted in other corn-growing areas to determine the distribution and extent of disease, and to identify the leafhopper vectors. In the meantime, the concerned agencies are responding proactively to prevent further spread by providing effective disease management recommendations.

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